

Analysis of edX Courses: Certification Rates Across Subjects

INTRODUCTION

edX, a platform developed by **MIT** and **Harvard**, provides access to **online courses** for millions of learners worldwide. However, not all courses achieve the same level of completion. This project investigates **how certification rates vary across different subjects** and institutions to better understand learner behavior and course effectiveness.



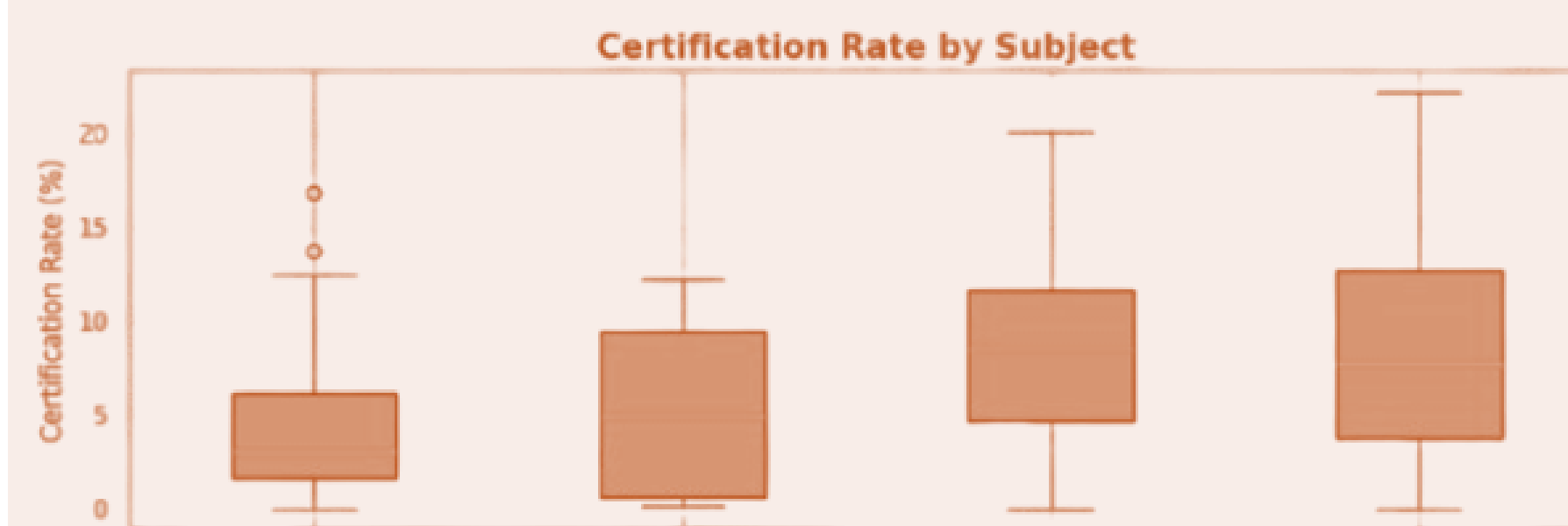
Research Question: Do certification rates differ significantly across different course subject areas?

Why It Matters:

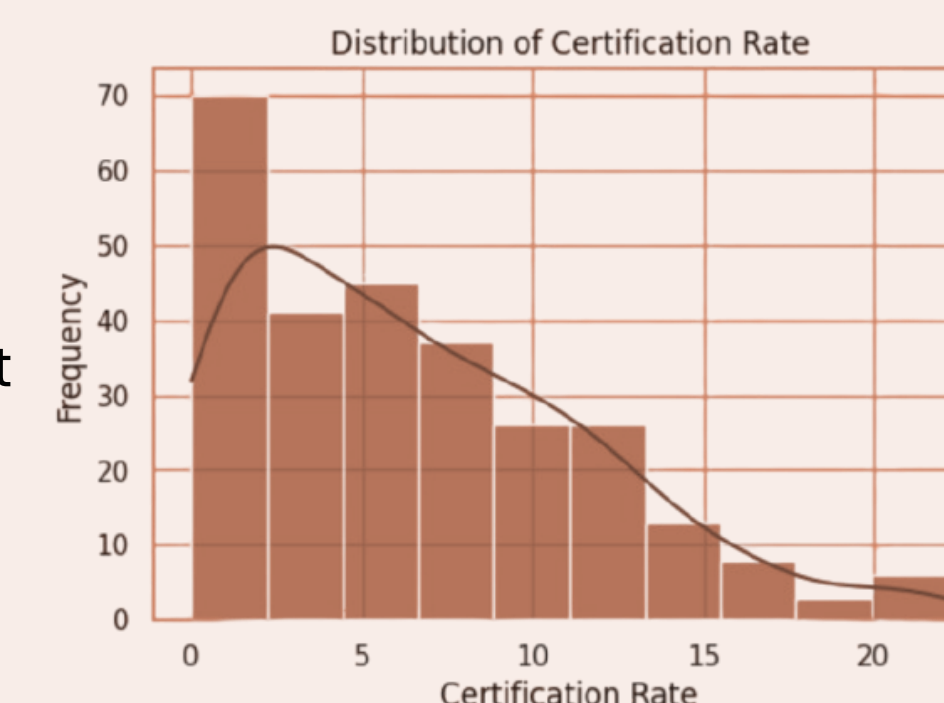
Identifying these differences helps improve course design, increase student engagement, and enhance overall learning outcomes.

Analysis

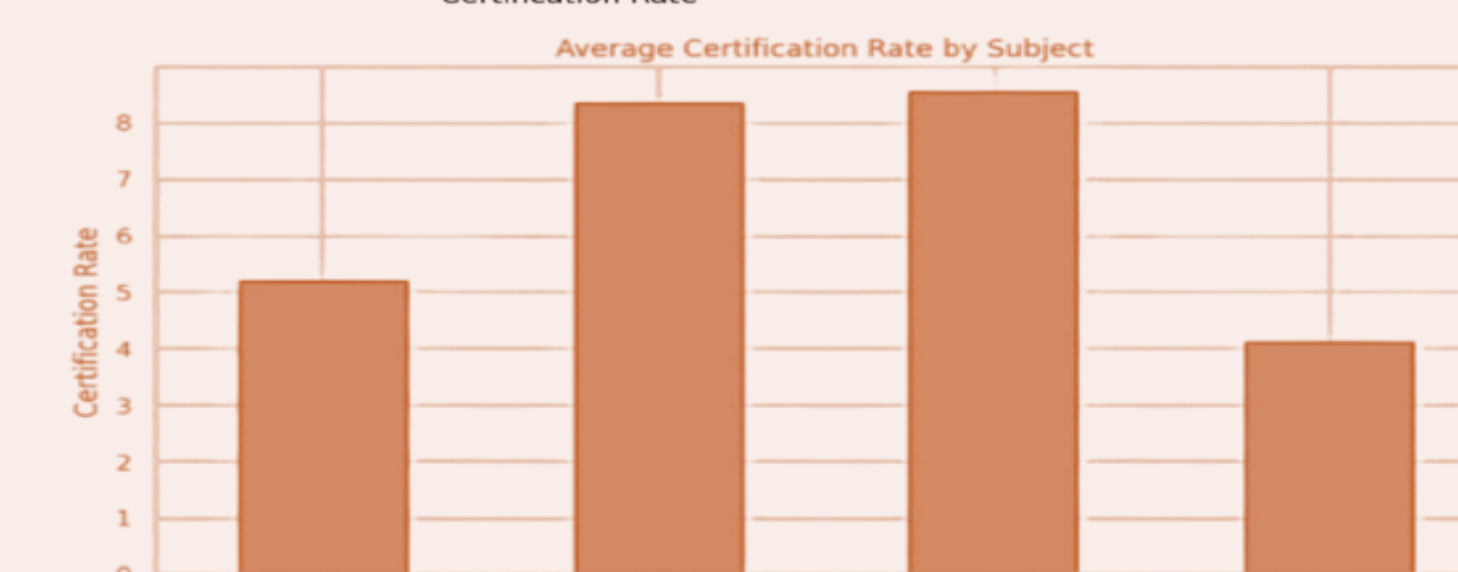
Certification rates are generally low across most courses, with clear variation between subjects. Some subjects achieve higher completion rates, while others remain consistently lower. The visualizations highlight differences in distribution, average values, and variability, indicating that **certification outcomes are influenced by subject area and course characteristics**.



Certification rates **differ across subjects**, with noticeable variation in median values and distribution within each field.



Distribution of certification rates shows that **most courses have low completion**, with a **small number achieving higher rates**.



Average certification rates vary across subjects, with **some fields achieving higher completion** than others.

Hypothesis

Certification rates in Computer Science courses may vary between institutions (MIT and Harvard).

- **H₀:** No significant difference in certification rates between MIT and Harvard (Computer Science)
- **H₁:** There is a significant difference in certification rates between MIT and Harvard courses in Computer Science.

Methodology



• Data Source & Dataset Description:

- HarvardX & MITx dataset (edX, 2012–2016)
- 290 courses, 4.5M participants, 250K certifications
- The data captures learner behavior, course structure, and engagement metrics such as video interaction and forum activity.

• Variables Analyzed:

- **Numerical:** Participants, Certified, % Certified
- **Categorical:** Course Subject, Institution
- **Engagement Metrics:** % Played Video, % Posted in Forum

• Analysis Approach:

- Compared certification rates across subjects
- Analyzed MIT vs Harvard differences
- Applied T-test for significance
- Used visualizations (Histogram, Bar Chart, Boxplot)

Conclusion

- Certification rates vary significantly across different course subjects, leading to the **rejection of the null hypothesis** that assumes no difference between institutions.
- The analysis shows that course subject plays a major role in learner completion, with some fields achieving higher certification rates than others.
- Differences observed between MIT and Harvard courses indicate that **institutional factors and course design** influence certification outcomes.
- The distribution of certification rates reveals that **most courses have relatively low completion**, while only a few achieve higher rates, highlighting variability across the dataset.
- Visualizations further confirm that **certification rates are not consistent within subjects**, showing variation in median values and spread, which suggests differences in course difficulty and learner engagement.
- **Implications:**
 - Improving course structure and content delivery can enhance completion rates
 - Increasing learner engagement may lead to better certification outcomes